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We would like to thank you for your contribution to the Scientific Programme at the 33rd International Congress of Psychology held July 21 2024 - July 26 2024.



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Validity of Computerized Neuropsychological Assessment in Healthy Middle-Aged and Elderly Koreans.

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Introduction

- Previous studies suggest that computerized neuropsychological assessments have many potential benefits in both clinical and research settings. Compared to traditional neuropsychological assessments, computerized tests can reduce administration time and costs, examiner error, and so on (Schatz & Brownwyke, 2002).
- The Cambridge Neuropsychological Test Automated Battery (CANTAB) is a widely used computerized assessment battery, across various countries, age groups, and clinical populations for assessing cognitive functions and previous studies verified its validity and reliability many times (Gonçalves, et al., 2017; Lenehan et al., 2016; Robbins et al., 1994, 1998; Smith et al., 2013).
- Researchers in Korea have employed CANTAB to assess cognitive function (Kang et al., 2023; Lee et al., 2021). However, empirical evidence for its validation in middle-aged and elderly Koreans remains limited.
- In this study, we aimed to evaluate test validity by comparing CANTAB tests with paper-pencil tests in healthy middle-aged and elderly Koreans.

Method

Procedure

❖ First visit

Informed consent, Background characteristics survey

❖ First visit

CANTAB tests

Traditional neuropsychological tests

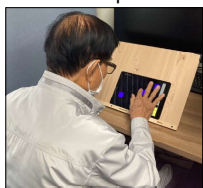
• 2~4 weeks

❖ Second visit

Traditional neuropsychological tests

CANTAB tests

❖ CANTAB process



- Participants followed instructions and carried out the tests by themselves.
- When participants failed to understand instructions, experimenter explained them.
- CANTAB tests were used on an APPLE iPad 9th, which display size of 10.2 inches and resolution of 2160 x 1620.

Results

Neuropsychological test battery

subset	M(SD)	Cognitive domain measured
Traditional battery		
Animal Naming	15.75 (4.62)	Verbal Fluency
Boston Naming Task	13.49 (1.21)	Verbal Ability
Word List Memory	19.83 (3.75)	Memory
Word List Recall		Memory
total counts	6.55 (2.19)	
storage (%)	78.88 (22.18)	
Word List Recognition	9.34 (1.09)	Memory
Constructional Praxis Recall	7.74 (2.91)	Memory
Trail Making Test Part A (ms)	48.69 (19.54)	Processing Speed
Trail Making Test Part B (ms)	113.89 (59.84)	Processing Speed
Controlled Oral Word Association	27.31 (10.62)	Verbal Fluency
Vocabulary ^a (raw)	30.51 (9.35)	Verbal Ability
CANTAB battery		
Verbal Recognition Memory (VRM)		Memory
free recall	4.69 (2.43)	
immediately recognition	30.60 (3.08)	
delayed recognition	29.52 (3.30)	
Pattern Recognition Memory (PRM)		Memory
immediately recognition (%)	92.18 (9.65)	
delayed recognition (%)	79.36 (15.32)	
Paired Associates Learning (PAL)		Memory
total errors	29.97 (16.86)	
first attempt memory score	8.85 (4.04)	
Rapid Visual Information Processing (RVP)		Attention / Preprocessing Speed
A' (sensitivity index)	.98 (.02)	
latency (ms)	484.20 (109.60)	
Multitasking Test (MTT)	301.88 (185.90)	Executive Function
Spatial Working Memory (SWM)	17.98 (8.70)	Memory

Note. ^aVocabulary is a subset of Korean Wechsler Adult Intelligence Scale-IV (K-WAIS-IV).

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Participants (N=65)

variables	M(SD)
Gender female, n (%)	43 (66.2)
Age, years	72.77 (7.17)
Education, years	12.65 (3.26)
CIST ^a	27.14 (1.86)
MMSE ^b	26.95 (1.85)
CERAD-K ^c	83.62 (10.68)
HADS-Anxiety ^d	3.52 (2.74)
HADS-Depression ^d	5.86 (3.47)

Note. Participants' ages ranged from 54-93 years.

^aCIST=Cognitive Impairment Screening Test, ^bMMSE=Mini-Mental State Examination, ^cCERAD-K= The Korean Version of the Consortium to Establish a Registry for Alzheimer's Disease the 2nd Edition,

^dHADS=Hospital Anxiety and Depression Scale

- Principal component analysis** of traditional neuropsychological subsets and **Comparison** of factors with individual subsets from CANTAB.

Traditional neuropsychological subset	1	2	3
Word List Recall	.94	.06	.07
Word List Recall (storage)	.93	-.01	.02
Word List Memory	.80	.08	.09
Word List Recognition	.72	.21	.30
Constructional Praxis Recall	.57	.41	.25
Trail Making Test Part B	-.17	-.83	-.08
Trail Making Test Part A	-.19	-.76	.26
Controlled Oral Word Association	.18	-.06	.77
Animal Naming	-.21	.59	.61
Vocabulary (raw)	.27	-.05	.60
Boston Naming Task	-.00	.47	.47
Eigenvalue	3.46	2.05	1.77
Variance (%)	31.44	18.63	16.11
KMO=.69, Bartlett's $\chi^2=267.65$, $df=55$, $p<.001$			
CANTAB subset	1	2	3
VRM free recall	.36**	-.12	.10
VRM immediately recognition	.42**	.01	.26*
VRM delayed recognition	.37**	.08	.31*
PRM % correct (immediately)	.39**	-.26*	.33**
PRM % correct (delayed)	.12	-.20	.23
PAL total errors	-.46***	.27*	-.47***
PAL first attempt memory score	.36**	-.27*	.44***
RVP latency	-.28*	.21	-.17
RVP A' (sensitivity index)	.10	-.21	.04
MTT multitasking score	.32**	-.09	.01
SWM between error	-.10	.26*	-.19

* $p<.05$, ** $p<.01$, *** $p<.001$

Conclusions

- We found some correlations between CANTAB subtests and traditional cognitive tests. In particular, CANTAB memory tests correlated with memory and verbal ability.
- The significant correlation of CANTAB tests with several factors is consistent with previous studies (Lenehan et al., 2016; Smith et al., 2013), indicating that the results of CANTAB measure multiple cognitive functions simultaneously.
- Previous studies lacked traditional processing speed and executive function tests. Therefore, we should include these tests and compare them with other CANTAB subtests.
- The sample size is too small to conduct factor analysis; however, we are currently recruiting more participants. If we collect enough data, we can use other statistical methods to analyze CANTAB's validity.
- Despite the insufficient sample size or tests, we found that memory and verbal ability tests are relevant. If these limitations are addressed, we could reveal more results validating computerized neuropsychological assessments in middle-aged and elderly Koreans.

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